

Paturi Bhavana

SUMMARY

Enthusiastic and detail-oriented Computer Science student with hands-on experience in full-stack web development, machine learning, and deep learning projects. Skilled in building responsive applications, database management, and developing AI-based solutions using modern technologies. Passionate about problem-solving, continuous learning, and contributing effectively in collaborative team environments.

PROJECTS

DecisionLoop AI AI-Powered Company Research & Recommendation System

Mar–Jun 2026

LLMs, Embeddings, RAG, PostgreSQL

- Developed an AI-powered decision-support system using LLMs, embeddings, RAG, and PostgreSQL to retrieve and rank company and market evidence and generate evidence-backed recommendations.
- Built an interactive dashboard with agent-based research workflows, displaying recommendations, supporting evidence, confidence scores, and evaluation metrics.

StreamSavvy OTT Streaming Platform

Nov 2025

React.js, Movie APIs, UI/UX

- Developed a fully responsive OTT streaming platform using React.js with dynamic routing and search functionality.
- Integrated external movie APIs to fetch and display real-time streaming content.
- Built reusable UI components with efficient state management and responsive design.

Synexa AI

Feb–May 2026

LLM APIs, Agent-Based Workflows, Embeddings, Semantic Search

- Built an AI-powered system using LLM APIs and agent-based workflows to process user queries and generate evidence-backed responses.
- Implemented semantic search, embeddings, and retrieval pipelines to identify and rank relevant information.
- Evaluated system performance through testing, response validation, latency analysis, and failure-case identification.

RESEARCH

Forest Fire Detection using YOLOv12

Aug 2025 – Jan 2026

Research Student, SRM University AP

- Completed research on real-time forest fire detection using YOLOv12 with improved precision and accuracy.
- Fine-tuned model parameters and optimized inference speed for early wildfire detection and rapid response.
- Designed a real-time monitoring framework suitable for large-scale forest surveillance systems.
- **Published at ICIPCN 2026**; available on IEEE Xplore: [View Paper](#).

Deep Metric Learning-Based Image Retrieval for Smart Agriculture

Jan 2026 – Jun 2026

Research Student, SRM University AP

- Developed a deep metric learning framework for agricultural image retrieval using a lightweight CNN and triplet loss.
- Improved retrieval accuracy by learning similarity-preserving image embeddings with online semi-hard triplet mining.
- Validated the framework on PlantVillage and AI Challenger datasets, achieving superior Precision@K and mAP over conventional CNN-based retrieval methods.
- **Published at IEEE InterSys 2026**; available on IEEE Xplore: [View Paper](#).

INTERNSHIP EXPERIENCE

IBM

Machine Learning Intern

– Developed a machine learning model to predict employee burnout rates based on multiple organizational and personal factors.

– Performed data preprocessing, feature analysis, and model evaluation to identify at-risk employees and propose data-driven support strategies.

Jun 2024 – Jul 2024

Infosys Springboard Internship 6.0

Project: EnviroScan AI-Powered Pollution Source Identifier

– Currently developing an AI-powered system to identify and analyze pollution sources using geospatial data.

– Leveraging geospatial analytics and machine learning techniques to detect pollution patterns and affected regions.

– Aiming to support environmental monitoring and data-driven decision-making for sustainable development.

Feb 2026 – may 2026

EDUCATION

SRM University AP

B.Tech in Computer Science Engineering

2023 – Present

CGPA: 8.73

Sri Chaitanya Junior College

Intermediate (MPC)

2021 – 2023

Percentage: 89%

CERTIFICATIONS

MongoDB Associate Developer Certification

MongoDB

– Learned MongoDB database design, CRUD operations, indexing, aggregation pipelines, and database management concepts.

2026

Java SE 17 Developer Certification

Oracle / Java Programming

– Gained knowledge in Java SE 17 concepts including OOP, collections, exception handling, multithreading, and Java application development.

2026

SKILLS

Languages: C, C++, Python, JavaScript
Web: HTML, CSS, React.js, Bootstrap
Backend: Node.js, PHP, FastAPI
Databases: MySQL, MongoDB, PostgreSQL
AI/ML: Machine Learning, Deep Learning, APIs, RAG, Embeddings, Vector Search, Semantic Search
Computer Vision: OpenCV, PyTorch
Tools: Git, GitHub, VS Code, Jupyter Notebook, MySQL Workbench, Roboflow, GitHub Copilot
Operating Systems: Linux, Windows
Soft Skills: Communication, Leadership, Teamwork, Problem Solving